BIRATIONAL MORPHISMS INDUCE MORPHISMS OF NORMALIZATIONS OVER A DENSE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Normalization over a dense open. lemma-finite-over-dense-open (carded L26752). The finite covers over dense opens.
- Birational maps of normal schemes. Morphisms, lemma birational-induced-morphism-normalizations (cross-chapter, named). The scheme case.

Source reference. Original lemma 0ABV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4792–4792. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BB9: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4911–4911. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ABT: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1421–1421. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BB4: focused-seeds/geometry/S9-decent-birational-induced-morphism-normalizations/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BB1: focused-seeds/geometry/S9-decent-birational-induced-morphism-normalizations/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a birational morphism between algebraic spaces over S which are decent and have finitely many irreducible*

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components. Then the normalizations $X^\nu \to X$ and $Y^\nu \to Y$ exist and there is a commutative diagram

$$\begin{array}{ccc} X^\nu & \longrightarrow & Y^\nu \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

of algebraic spaces over S. The morphism $X^\nu \to Y^\nu$ is birational.

Proof. By Lemma 0BB9 we see that X and Y satisfy the equivalent conditions of Morphisms of Spaces, Lemma 0BB1 and the normalizations are defined. By Morphisms of Spaces, Lemma 0BB4 the algebraic space X^ν is normal and maps codimension 0 points to codimension 0 points. Since f maps codimension 0 points to codimension 0 points (this is the same as generic points on decent spaces by Lemma 0ABV) we obtain from Morphisms of Spaces, Lemma 0BB4 a factorization of the composition $X^\nu \to X \to Y$ through Y^ν.

Observe that X^ν and Y^ν are decent for example by Lemma 0ABT. Moreover the maps $X^\nu \to X$ and $Y^\nu \to Y$ induce bijections on irreducible components (see references above) hence X^ν and Y^ν both have a finite number of irreducible components and the map $X^\nu \to Y^\nu$ induces a bijection between their generic points. To prove that $X^\nu \to Y^\nu$ is birational, it therefore suffices to show it induces an isomorphism on local rings at these points. To do this we may replace X and Y by open neighbourhoods of their generic points, hence we may assume X and Y are affine irreducible schemes with generic points x and y. Since f is birational the map $\mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$ is an isomorphism. Let $x^\nu \in X^\nu$ and $y^\nu \in Y^\nu$ be the points lying over x and y. By construction of the normalization we see that $\mathcal{O}_{X^\nu,x^\nu} = \mathcal{O}_{X,x}/\mathfrak{m}_x$ and similarly on Y. Thus the map $\mathcal{O}_{X^\nu,x^\nu} \to \mathcal{O}_{Y^\nu,y^\nu}$ is an isomorphism as well. $\square$